

Chapter ~ 3

Electric Resistance & Ohm's law

=> Electric Current

Electric Current refers to the continuous and directed flow of electric charges.

=> Types : DC Current: Current flows in one direction only, Magnitude is constant.

AC Current: Current direction changes periodically, Magnitude also changes.

$$I = \frac{Q}{T} \rightarrow \begin{matrix} \text{Charge} \\ \text{Time} \end{matrix} \quad (Q = IT)$$

=> Current Density (j)

$$j = \frac{I}{A} \rightarrow \begin{matrix} \text{Current} \\ \text{Area} \end{matrix}$$

Unit : A/m^2

=> Resistance (R)

The ratio of the potential difference to the electric current is known as resistance.

$$V \propto I$$

$$V = IR.$$

* (temp should be constant)

Unit is ohm (Ω)

Dimensions : $[M^1 L^2 T^{-3} A^{-2}]$

$\Rightarrow$ Conductance (G)

$$G = \frac{1}{R}$$

Unit : Ω^{-1} or S (Siemens)

Dimensions : $[M^{-1} L^{-2} T^3 A^2]$

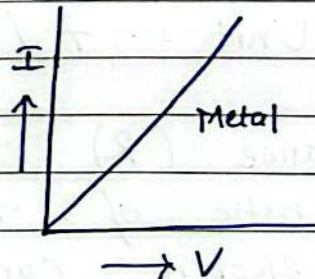
$\Rightarrow$ OHM'S LAW

$$\frac{V}{I} = \text{constant}$$

$$\frac{V}{I} = R \Rightarrow \boxed{V = IR}$$

Acc to this law, If there is no change in the physical state of a conductor (such as temperature) then the ratio of the potential difference applied at its ends and the current flowing through it is constant.

Graph V vs I



$\Rightarrow$ Flow of Electrons

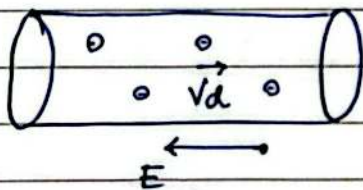
i) Relaxation time.

Free electrons do not sit still, they crash into ions, the average time that passes between one collision and the next is called relaxation time.

ii) Drift velocity

It is the slow, steady net speed electrons gain when pushed by an electric field.

$\Rightarrow$ Current, Drift velocity Relation ($J = nev_d$) or ($I = n e A v_d$)



$$I = \frac{Q}{T}$$

$$I = \frac{\text{no of electrons} \times e}{t}$$

$$\frac{\text{no of electrons}}{\text{volume}} = n$$

$$I = n \times A \times v_d \times l \times e$$

$$\begin{aligned} \text{no of } e^- \text{'s} &= n \times \text{volume} \\ &= n \times A \times l \\ &= n \times A \times v_d \times t \end{aligned}$$

$$I = n A v_d e$$

$$\frac{I}{A} = n v_d e$$

$$J = n v_d e$$

⇒ Mobility of Electrons (μ_e)

drift velocity $V_d = a \tau$ — relaxation time
 acceleration

$$V_d = \frac{F}{m} \tau \quad (F = ma)$$

$$V_d = \frac{eE}{m} \tau \quad (F = eE)$$

$$V_d = \frac{e \tau}{m} E$$

—————→ mobility

$$V_d = \mu_e E$$

$$V_d \propto E$$

$$\boxed{\mu_e = \frac{e \tau}{m}}$$

$$V_d = \mu_e E$$

$$\frac{V_d}{E} = \mu_e$$

$$\frac{V_d}{E} = \frac{J}{ne}$$

$$\boxed{\mu_e = \frac{J}{ne}}$$

$$\text{or } \boxed{\mu_e = \frac{V_d}{E}}$$

⇒ Derivation of Ohm's law.

We know that,

$$J = neV_d$$

$$I = neAV_d$$

$$\frac{I}{neA} = V_d$$

$$\frac{I}{neA} = aT \quad (V_d = aT)$$

$$\frac{I}{neA} = \frac{F}{m} T \quad (a = F/m)$$

$$\frac{I}{neA} = \frac{eE}{m} T \quad (F = eE)$$

$$\frac{I}{neA} = \frac{eV}{ml} T \quad (E = \frac{V}{l})$$

$$I = \frac{ne^2}{m} AE T$$

$$I = \frac{AE}{\rho}$$

$$\frac{I}{A} = \frac{E}{\rho}$$

$$J = \sigma E \quad (\rho = 1/\sigma)$$

$$\frac{Iml}{eT} = VneA$$

$$\frac{Iml}{ne^2 TA} = V \quad R$$

$$I \propto V$$

$$R = \frac{ml}{ne^2 TA}$$

⇒ Resistivity

$$R \propto l$$

$$R \propto \frac{l}{A} \Rightarrow R = \frac{\rho l}{A}$$

$$\text{Resistivity} \Rightarrow \rho = \frac{m}{ne^2 T}$$

⇒ Dependence of Resistance on Temperature

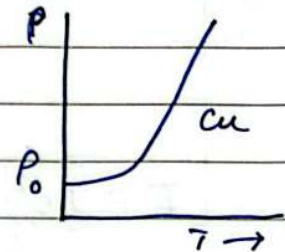
$$R_2 = R_1 [1 + \alpha t] \quad \text{or} \quad R_2 = R_1 [1 + \alpha (t_2 - t_1)]$$

$$R \propto T$$

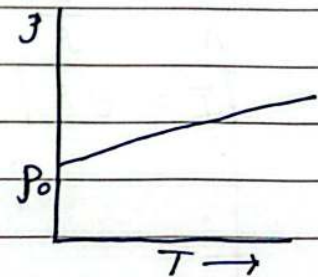
$$\therefore R_2 = R_1 [1 + \alpha (T_2 - T_1)]$$

⇒ Resistivity of Materials

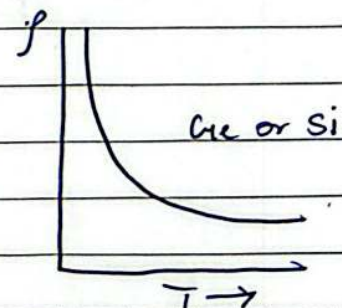
i) Resistivity of Metals: For several metals resistance (or resistivity) is nearly proportional to absolute temperature.



ii) Resistivity of Alloys: It also increases with rise in temperature but it is much smaller



iii) Resistivity of Semiconductors: There are certain substances whose electric conductance is large its decreases



=> Specific Resistance

$$\rho = \frac{E}{J}$$

or

$$\rho = R \left(\frac{A}{l} \right)$$

=> Combination of Resistances

i) In Series:

$$R = R_1 + R_2 + R_3 \dots$$

ii) In Parallel:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$

=> Electric Power

Rate at which electric energy is transferred into other forms of energy is called electric power

$$P = W/t \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$$

Units: KW-h

$$1 \text{ KW-h} = 3.6 \times 10^6 \text{ J}$$

=> Specific Resistance is the ratio of intensity of electric field E and current-density at that point.

$\frac{E}{J}$ - electric field.
 J - current density

Dimensions: $[M L^3 T^{-3} A^{-2}]$

⇒ Electric Cell

An Electric Cell is a source of electrical energy which maintains a continuous flow of charge in a circuit.

the cell changes chemical energy into electrical energy.

⇒ Electromotive force of a Cell (EMF) = V

The work done by the cell in forcing unit positive charge to flow in the whole circuit is called the electromotive force of the cell.

⇒ Terminal potential difference of a cell is equal to work done for the flow of unit charge in the external circuit

$$V = \frac{W_{ext}}{q}$$

Internal Resistance of a Cell.

⇒ The resistance offered by the electrolyte of the cell to the flow of current through it is called internal resistance of the cell.

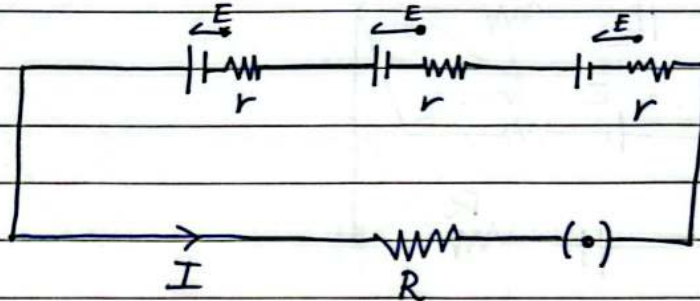
It depends upon

- i] Directly proportional to the separation between two plates of the cell.
- ii] Inversely proportional to the plates area dipped into electrolyte
- iii] Directly proportional to concentration
- iv] Inversely proportional to temperature

⇒ Combinations of cells.

* 1. In Series

⇒ Series combination of cells of same EMF



$$E_{\text{net}} = E + E + E = nE.$$

$$R_{\text{eq}} = R + nr = R + nr$$

$$I = \frac{nE}{R + nr}$$

Special case

i] If $nr \ll R$

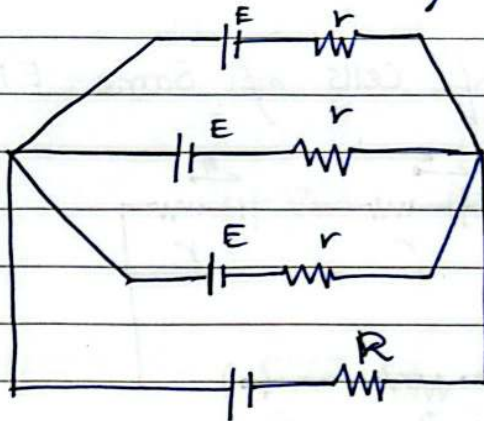
$$\Rightarrow I = \frac{nE}{R}$$

ii] If $nr \gg R$

$$\Rightarrow I = \frac{nE}{nr} = \frac{E}{n}$$

* 2. In Parallel

⇒ Parallel combination of cells of same EMF.



$$E = E$$

$$R_{eq} = \left(\frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \dots \right)^{-1} + R.$$

$$= \frac{r}{n} + R.$$

Now, $I = \frac{E}{\frac{r}{n} + R}$

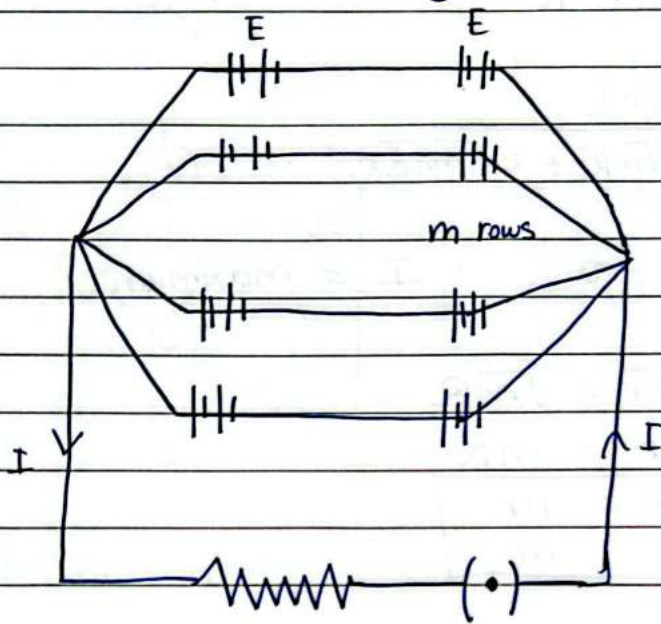
$$I = \frac{nE}{r + nR}$$

Special Case

i] If $r \ll R$ $I = \frac{E}{R}$

ii] If $r \gg R$ $I = \frac{nE}{r}$

* 3. In Mixed Cirraping



$$E = nE$$

$$R_{eq} = \left(\frac{1}{nr} + \frac{1}{nr} + \frac{1}{nr} + \dots m \right)^{-1} + R.$$

$$= \left(\frac{m}{nr} \right)^{-1} + R$$

$$= \frac{nr}{m} + R$$

Now, $I = \frac{nE}{\frac{nr}{m} + R}$

$$I = \frac{mnE}{nr + mr}$$

$$I = \frac{mnE}{\sqrt{nr} - \sqrt{mR}}$$

$$I = \frac{mnE}{(\sqrt{nr} - \sqrt{mR})^2 + 2\sqrt{mnRr}}$$

When $\sqrt{nr} - \sqrt{mR} = 0$, $I = \text{maximum}$.

$$\sqrt{nr} = \sqrt{mR}$$

$$nr = mR$$

$$R = \frac{nr}{m}$$

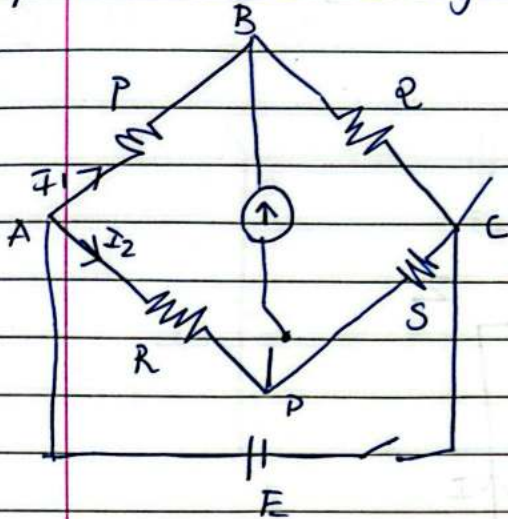
$\Rightarrow$ Kirchhoff's law

- a) Kirchhoff's current law (KCL) : At any junction in an electrical circuit, the algebraic sum of currents entering junction is equal to sum of currents leaving the junction.
- Conservation of electrical charge.

- b) Kirchhoff's Voltage law (KVL) : The algebraic sum of potential differences in a closed loop is zero.

Conservation of Energy

⇒ Wheatstone Bridge.



$$\frac{P}{Q} = \frac{R}{S}$$

⇒ Potentiometer

Sensitivity: when the jockey is slightly displaced from the null point position, the galvanometer shows a large deflection, then it is said to be sensitive.

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

i) Comparison of emf's of two cells

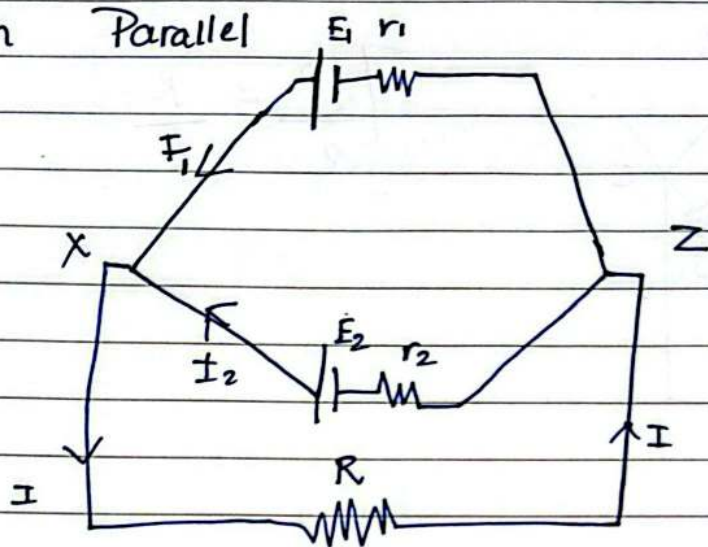
$$r = R \left[\frac{l_1}{l_2} - 1 \right]$$

ii) Determination of internal resistance of Primary cell.

⇒ Note: When we measure emf of a cell by potentiometer then in the position of null point no current flows, hence we obtain the actual value of emf of the cell.

⇒ Combinations of Cells. for unequal EMF.

i) In Parallel



$$[V = E - Ir]$$

Current from battery $\Rightarrow I = I_1 + I_2$.

$$V = E_1 - I_1 r_1 \Rightarrow I_1 = \frac{E_1 - V}{r_1}$$

$$V = E_2 - I_2 r_2 \Rightarrow I_2 = \frac{E_2 - V}{r_2}$$

$$I = \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$$

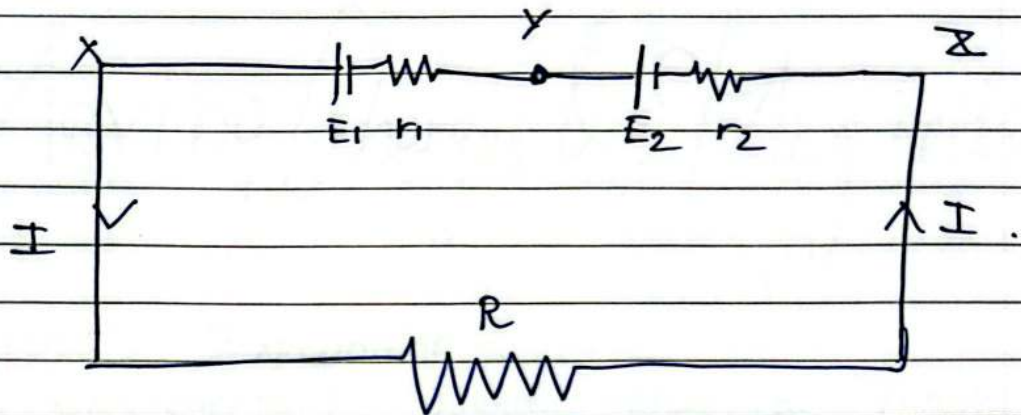
$$V = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right] - I \left[\frac{r_1 r_2}{r_1 + r_2} \right]$$

as $V = E - IR$.

$$E_{eq} = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right]$$

$$R_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

⇒ In Series.



$$\begin{aligned} V_{xy} &= V_x - V_y \\ &= E_1 - I r_1 \end{aligned}$$

$$\begin{aligned} V_{yz} &= V_y - V_z \\ &= E_2 - I r_2 \end{aligned}$$

$$\begin{aligned} \text{Total } V_{eq} &= V_{xy} + V_{yz} \\ &= (E_1 - I r_1) + (E_2 - I r_2) \end{aligned}$$

$$E_{eq} - I r_{eq} = (E_1 + E_2) - I (r_1 + r_2)$$

$$E_{eq} = E_1 + E_2$$

$$r_{eq} = r_1 + r_2$$